INVERT LAB

DOTS Sprite Animator

User Guide

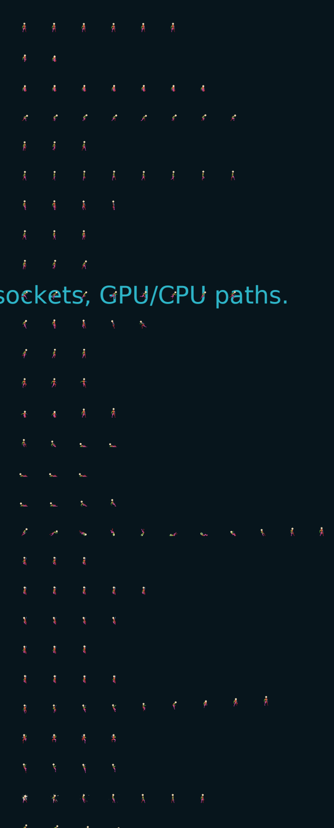
Version 0.7.1 · 27 August 2026 · Unity 6000.5

WINDOW	Window > DOTS Sprite Animator
PACKAGE	com.invertlab.spriteanimator
NAMESPACE	InvertLab.Sprites.DOTS
EDITOR TITLE	DOTS Sprite Animator 0.7.1
HEADER LINE	v0.7.1 · DOTS AUTHORING STUDIO

The live editor constant PackageVersion is "0.7.1" (window title and toolbar). package.json is still 0.7.0.

Authoring studio for DOTS-first 2D flipbook animation — clips, events, colliders, sockets, GPU/CPU paths.

Showcase sheet · sword.png



Showcase sheet · hero.png

1 · What is in this guide

This manual documents Invert Lab DOTS Sprite Animator as it ships in the live editor window titled DOTS Sprite Animator 0.7.1. The toolbar header reads SPRITE ANIMATOR with the muted line v0.7.1 · DOTS AUTHORING STUDIO. Open it from Window > DOTS Sprite Animator.

Field tables quote the exact GUIContent labels from SpriteSheetToolWindow.cs. Defaults and ranges come from SpriteSheetProfile.cs. This is not the older 0.5 BallForge tutorial: clip tint, flip, texture filter, clip-list drag reorder, and inspector foldouts are not in this window.

> **About the figures**
> The window, preview, timeline, and inspector diagrams in this guide are labeled illustrations of the v0.7.1 editor. They are not live Unity Editor screenshots. Showcase sheets (hero.png, sword.png) are sample flipbook textures, not editor chrome.

Product identity

Item	Detail
Publisher	Invert Lab (package author field: InvertLab)
Product	DOTS Sprite Animator
Package id	com.invertlab.spriteanimator (internal; not shown on the store)
Runtime namespace	InvertLab.Sprites.DOTS
Editor namespace	InvertLab.Sprites.DOTS.Editor
Window class	SpriteSheetToolWindow (EditorWindow, min size 860 × 610)
Window title	DOTS Sprite Animator 0.7.1
Menu	Window > DOTS Sprite Animator
Tools menu	Tools > DOTS Sprite Animator > Validate Installation / Help / Open Window
Editor version constant	PackageVersion = 0.7.1 (window title and toolbar)
package.json version	0.7.0 (mismatch with the editor header — document both)
Unity	6000.5 (package.json unity field is 6000.0; open with Unity 6000.5)
Guide date	27 August 2026
0.7.1 addition	Toolbar List opens the floating Undo / Redo history panel (sits beside Redo).

Chapters

Work through the studio left-to-right, then down: clips, preview, inspector, timeline. Runtime and GPU rules close the book. Every inspector group is an always-visible SectionLabel. Tables use the columns Control · Type / default · What it does.

2 · Install and open

Embed or reference the package at Packages/com.invertlab.spriteanimator. The studio is a portable Unity 6000 package: Entities, Entities Graphics, and URP only — no NZCore, Rukhanka, Trove, or project-only assemblies.

Required packages

Control	Type / default	What it does
com.unity.entities	UPM dependency · 1.4.2 in package.json	DOTS entity runtime. Checked by Validate Installation.
com.unity.entities.graphics	UPM dependency · 1.4.16 in package.json	Entities Graphics used by the sprite render init path.
com.unity.render-pipelines.universal	UPM dependency · 17.5.0 in package.json	URP. Shader lookup is part of validation (SpriteShaderLibrary.TryFindAll).

Menus

Control	Type / default	What it does
Window > DOTS Sprite Animator	MenuItem · always	Opens or focuses SpriteSheetToolWindow. Sets titleContent to "DOTS Sprite Animator " + PackageVersion (0.7.1). If the Project selection is a ScriptableSpriteSheetProfile, that asset loads on enable.
Tools > DOTS Sprite Animator > Open Window	Compatibility redirect	Calls the same Open(). Kept for projects that used the original tools-menu path.
Tools > DOTS Sprite Animator > Validate Installation	MenuItem · dialog + Console	Checks the three UPM packages and the shader library. Success dialog: DOTS Sprite Animator — Validation Succeeded. Failure: DOTS Sprite Animator — Validation Failed. Button OK. Logs with Debug.Log / Debug.LogError.
Tools > DOTS Sprite Animator > Help	MenuItem	Opens Documentation~/QuickStart.md if present, else package README.md, else a file:// URL to README.
Toolbar Check	Button · tooltip "Validate package dependencies and shader setup."	Same as Validate Installation.
Toolbar Help	Button · tooltip "Open DOTS Sprite Animator quick start docs."	Same as Tools > ... > Help.

After import
Run Tools > DOTS Sprite Animator > Validate Installation once. The dialog reports [OK] package-id (version) or [MISSING] package-id, plus the shader search result. Fix missing packages in the Unity Package Manager before authoring.

3 · Window layout and five-minute workflow

OnGUI paints a dark studio in a fixed chrome: a 48 px toolbar, a three-column work area, and a full-width timeline along the bottom. Default clip pane 210 px (min 180), inspector 340 px (min 260), preview takes the remainder (min 220). Timeline height is the smaller of 226 px and 38% of the window. An 8 px gap separates the toolbar from the work area.

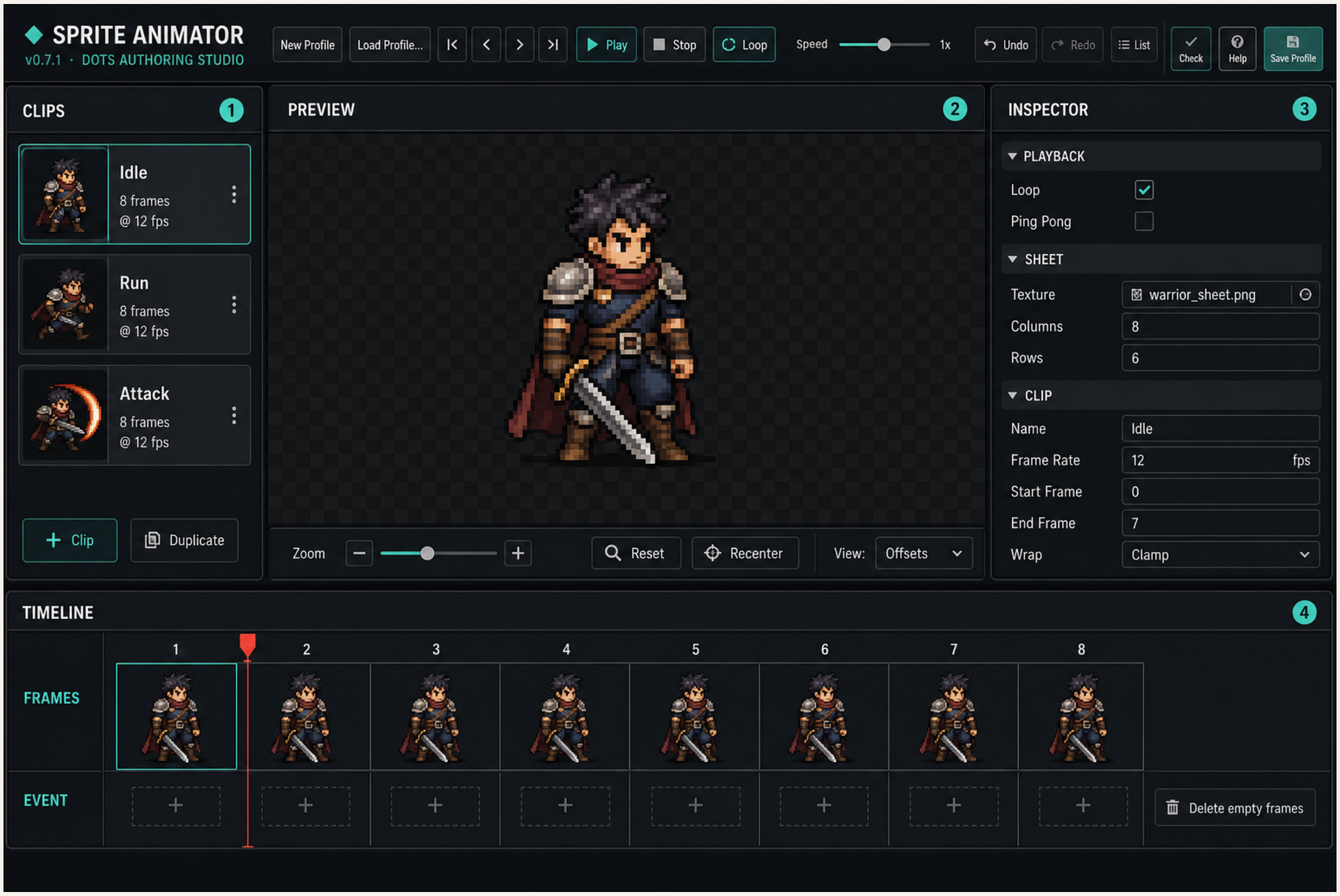


Figure 1.

Full window layout of DOTS Sprite Animator v0.7.1. Callouts mark (1) CLIPS, (2) PREVIEW, (3) INSPECTOR, and (4) TIMELINE under the transport toolbar. Splitters between the three columns are 8 px grips; double-click a splitter to reset that pane to its default width. Labeled illustration of the v0.7.1 editor — not a live Unity screenshot.

Chrome

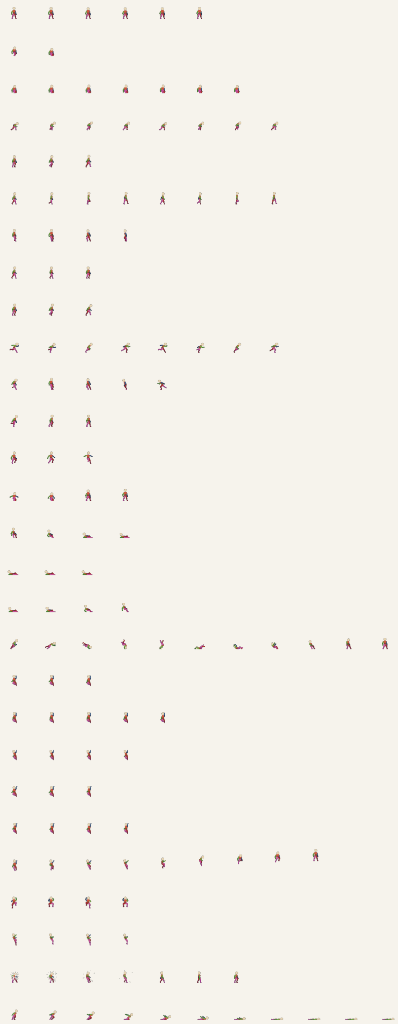
Control	Type / default	What it does
Toolbar	48 px strip	Branding, New/Load Profile, transport |< < > >|, Play/Pause, Stop, Loop, Speed 0.1–3, 1x, Undo, Redo, List, Check, Help, Save Profile, status text.
CLIPS pane	Default 210 px · min 180	Clip cards, + Clip / Duplicate / Delete. Left splitter grows this pane to the right.
PREVIEW pane	Remaining width · min 220	Zoom, Reset, Recenter, View: Offsets / View: Centered, canvas gizmos.
INSPECTOR pane	Default 340 px · min 260	Always-visible SectionLabel groups. Right splitter grows this pane to the left.
TIMELINE pane	Up to 226 px, full width	Ruler, EVENT lane, frame cards, red playhead, Delete empty frames.
Left / right splitters	8 px vertical grips · accent cyan on hover	Drag to resize. Double-click resets (“Reset clip panel width” / “Reset inspector panel width”).
Undo / Redo overlay	Floating GUI.Window · opened by List	Not docked. Title bar “Undo / Redo”. Drag the title bar to reposition.

Five-minute first profile

From a blank project that already has Entities, Entities Graphics, and URP:

1. Window > DOTS Sprite Animator. Confirm the header reads v0.7.1 · DOTS AUTHORING STUDIO.
2. Click New Profile. Status: Created new profile. Playback is paused.
3. In INSPECTOR · SHEET, assign Texture. File name, Size, and Cell size fill in. If a sibling <SheetName>_profile.asset exists, it auto-loads.
4. Set Columns and Rows (default 4 × 4) or click Auto-detect transparent grid.
5. Click + Clip. Rename with F2 (click only selects). Set Sheet Row, Frame Rate (8), Wrap Mode.
6. Scrub the timeline, press Space to preview, author events / colliders / sockets as needed.
7. Click Save Profile (enabled only when a sheet is assigned). Writes <SheetName>_profile.asset and matching .json beside the texture.

> **Showcase sheets**
> hero.png is a tall multi-row character sheet; sword.png is a compact attack sheet with smear frames. Drop either into Texture to walk the rest of this guide. Cell size is integer division of texture size by Columns × Rows — it is read-only.



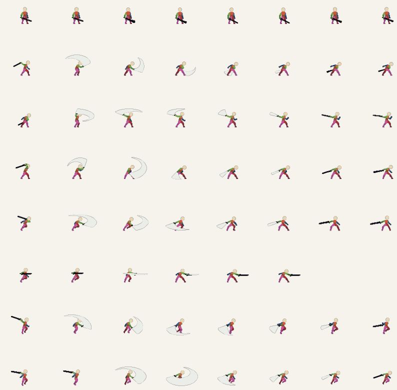


Figure 2.

Sample flipbook textures used throughout this guide: hero.png (left, multi-row locomotion set) and sword.png (right, compact smear-frame attacks). These are showcase sheets, not editor chrome. Not live Unity screenshots.

4 · Toolbar

The toolbar is a 48 px strip, slightly darker than the window. Buttons use the 11 pt, 28 px transport style. Labels below are exact GUIContent text. List was added in 0.7.1 and sits immediately beside Redo. The header always shows v0.7.1.

Control	Type / default	What it does
SPRITE ANIMATOR	Brand label	Left-side product mark (diamond prefix in the live window). No tooltip.
v0.7.1 · DOTS AUTHORING STUDIO	Muted version line	From PackageVersion. This is the string to trust for the running editor, not package.json.
New Profile	Button · tooltip “Create a fresh in-memory profile.”	Drops the bound asset, allocates a new SpriteSheetProfile (4×4, PPU 100, pivot 0.5/0.5, onion off, two default event defs, no clips). Resets selection and time, pauses. Status: Created new profile.
Load Profile…	Button · tooltip “Load a ScriptableSpriteSheetProfile asset.”	Native picker titled “Load DOTS Sprite Animator Profile”, filter asset, starts in Application.dataPath. Must live inside the Unity project and be a ScriptableSpriteSheetProfile. Notification: Loaded {name}.
|<	Button · tooltip “Jump to first frame.” · disabled with no clip	StepToBoundary(forward: false). Pauses. Status: Jumped to first frame.
<	Button · tooltip “Step one frame backward.” · disabled with no clip	StepFrame(−1). Wraps except Wrap Mode Once (clamped). Status: Stepped to frame N.
>	Button · tooltip “Step one frame forward.” · disabled with no clip	StepFrame(+1). Same wrap rules as the back-step button.
>|	Button · tooltip “Jump to last frame.” · disabled with no clip	StepToBoundary(forward: true). Status: Jumped to last frame.
Play / Pause	Button · disabled with no clip · default paused	Toggles playback. Play tooltip: “Play preview playback (Space).” Pause tooltip: “Pause preview playback.” Play is disabled when no clip is available (since 0.5.1).
Stop	Button · tooltip “Stop playback and return to time 0.”	Sets playing false and preview time to 0.
Loop	Toggle · default on	Tooltip: “Loop preview playback.” When off, preview stops when EvaluatePreview reports Ended.
Speed	Muted label + unlabeled horizontal slider · default 1.0 · range 0.1–3.0	Sets playback speed. Tick uses max(0.05, speed). Readout {n.n}x (for example 1.0x).
{n.n}x	Muted readout	Live speed value next to the slider.
1x	Button · tooltip “Reset preview speed to its 1x default.”	Disabled when already 1.0 (DefaultPreviewSpeed). Sets speed to 1.
Undo	Button · tooltip “Undo (Ctrl/Cmd+Z)”	Undo.PerformUndo(). Plain text (Windows-safe; no icon glyphs).
Redo	Button · tooltip “Redo (Ctrl/Cmd+Shift+Z or Ctrl+Y)”	Undo.PerformRedo().
List / Hide	Button · new in 0.7.1 · tooltip “Show the undo/redo history panel.”	Sits beside Redo. Label is List when the panel is closed, Hide when open. Toggles the floating Undo / Redo window. See chapter 18.
Check	Button · tooltip “Validate package dependencies and shader setup.”	Same as Tools > DOTS Sprite Animator > Validate Installation.

Control	Type / default	What it does
Help	Button · tooltip “Open DOTS Sprite Animator quick start docs.”	Same as Tools > DOTS Sprite Animator > Help.
Save Profile	Button · disabled until a sheet is assigned	Tooltip: “Save to <SheetName>_profile.asset and matching json.” See chapter 20.
(status text)	Muted label · default “Choose a sprite sheet to begin”	Live status between the Undo cluster and Save. The Undo/Redo/List cluster slides left if it would collide with Check.

Transport is clip-gated

|< < > >| and Play disable as a group when there is no current clip. Stop, Loop, Speed, Undo, Redo, List, Check, and Help remain available. Save Profile additionally requires a Texture.

5 · CLIPS

Header: CLIPS. Subtitle: {count} animation clips. A new profile starts with zero clips — click + Clip. There is no drag-reorder of clips (only timeline frames reorder, via Alt+drag on the thumbnail). Click selects; F2 is the only rename binding.

Control	Type / default	What it does
Clip card name	Bold label · tooltip “Select this clip. Press F2 to rename.”	Click selects (does not rename). Empty names display as Clip {i+1}. Selecting a different clip resets frame to 0, preview time to 0, and clears collider / event / onion selection.
Clip meta line	Muted · “{n} frames {fps:F1} fps”	Frame count and clip Frame Rate.
×	Mini-button · tooltip “Delete {clip.Name}. Undo is supported.”	Deletes that clip. Same undo as the footer Delete button.
+ Clip	Button · always enabled	Adds SpriteClipDef named Clip {count+1}, Row = count % Rows, frames = one sequential column per sheet column. Undo: Add Sprite Animation Clip.
Duplicate	Button · disabled with no current clip	Deep-copies the selection after it, named {Name} Copy (not uniquified). Copies frames, fps, wrap, durations, events, offsets, scales, rotations, tweens, facing, sockets. Does not copy hitboxes (those live on the profile keyed by clip name). Undo: Duplicate Sprite Animation Clip.
Delete	Button · disabled with no current clip	Deletes the selected clip and all hitboxes whose ClipName matches. Undo: Delete Sprite Animation Clip. Status: Deleted clip {name}.
F2 rename	Keyboard · clip list only	No double-click rename. Enter / keypad Enter commits; Escape cancels (Kept clip name {original}); click outside the name field commits. Names are trimmed and uniquified case-insensitively (suffix “ 2”, “ 3”, …). Undo: Rename Sprite Animation Clip. Hitboxes retarget via RenameHitboxClip.

Data defaults on new SpriteClipDef(): Name = Idle, Frames = {0,1,2,3}, Frame Rate = 8, Wrap Mode = Loop. + Clip overrides name and frames as above.

6 · PREVIEW

Header: PREVIEW. Empty canvas: Drop a sprite sheet in the Inspector. Checkerboard background, 18 px cells. Subtitle shows Frame {n}/{count} • {time:F2}s plus either offset {x, y} px or centered view, and optional suffixes for an armed collider tool, the socket tool, or polygon vertex count.

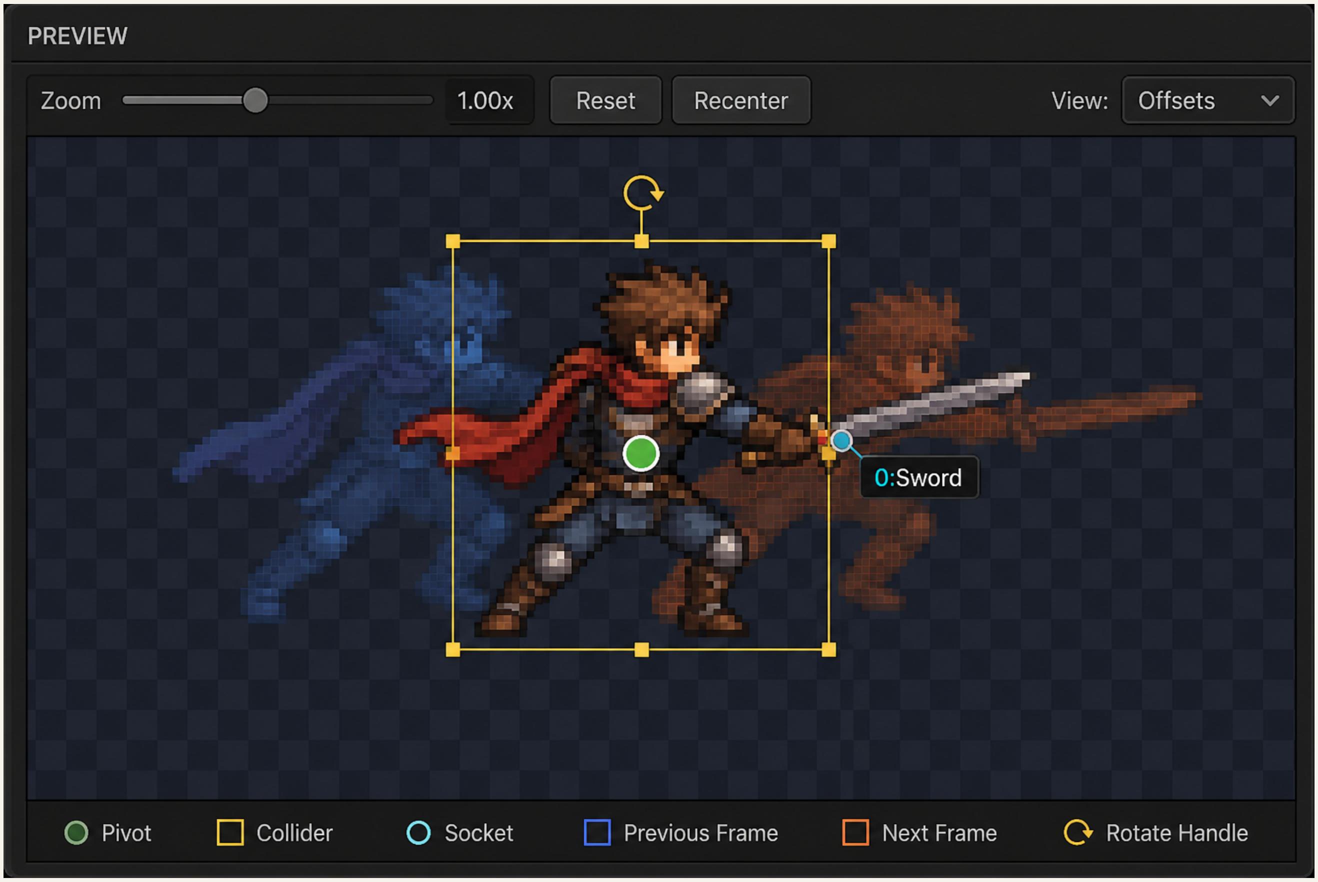


Figure 3.
PREVIEW pane of v0.7.1: Zoom 0.25–8, Reset, Recenter, View: Offsets / View: Centered. The canvas shows the green pivot disc, collider handles (move / scale / rotate), a click-placed socket, and onion-skin ghosts. Marquee selection uses the full canvas, not just the sprite cell. Labeled illustration of the v0.7.1 editor — not a live Unity screenshot.

Header controls

Control	Type / default	What it does
Zoom	Muted label + slider · default 1.00 · range 0.25–8.00	Preview magnification. Readout {zoom:F2}x. Mouse wheel over the canvas zooms toward the cursor (clamp(zoom × (1 − delta.y × 0.08), 0.25, 8)).
Reset	Mini-button · tooltip “Reset preview zoom and pan.”	Enabled when zoom is not 1 or pan/scroll is not 0. Sets zoom 1, pan 0, scroll 0. Status: Reset preview zoom and pan.
Recenter	Mini-button · tooltip “Keep zoom and center the sprite in the preview pane.”	Disabled if no sheet or already centered. Status: Recentered preview.
View: Offsets / View: Centered	Mini-button toggle · default View: Offsets (Authored)	Tooltip: “Toggle between authored per-frame playback offsets and centered source cells.” Same data as onion Playback Preview (Authored / Centered).

Navigation, pivot, marquee, gizmos

Control	Type / default	What it does
Scroll zoom	Mouse wheel over canvas	Zooms 0.25–8 toward the cursor. Scrollbars appear when zoomed content exceeds the pane.
MMB / Alt pan	Middle-mouse drag, or Alt+left-drag	Pans the preview. Alt+drag on the timeline thumbnail instead reorders frames — same modifier, different pane.
Show Pivot	Inspector toggle · default on	Draws a green disc at the sheet pivot (y-up UV). Drag it to author Pivot (clamped 0–1). Cursor MoveArrow. Undo: Move Sprite Pivot. Hidden pivot cannot be dragged.
Marquee on full canvas	Left-drag empty canvas	Cyan marquee. Shift / Ctrl / Cmd = additive. Selects visible (non-hidden) colliders whose AABB overlaps, and sockets on this frame whose label AABB overlaps. Creation tools own the canvas and suppress the marquee.
Collider handles	Gizmo when exactly one collider is selected	Body drag moves (multi-select moves all). Corner knobs scale from the opposite corner (8 px). Edge knobs scale on that axis. Rotate handle sits 26 px above the top edge and writes Angle (deg). Escape restores the start pose. Hidden colliders are not drawn unless selected; they still bake.
Click-to-place sockets	Armed by Add Socket	Button becomes “Click Preview to Place…”. Click the frame to create identity Socket {n} on the current frame (px, +x right, +y up, angle 0). Escape or right-click cancels. Same name is one identity across frames.

Ctrl+A selects preview objects, not timeline frames
Ctrl/Cmd+A selects all colliders (if Show Colliders is on) and all sockets on the current frame. It does not select every timeline card.

7 · TIMELINE

Header: TIMELINE. Empty state: Add a clip to build its timeline. Status line template: {n} frames • {total:F3}s • drag = marquee • Alt+drag image = reorder • frame edge = duration • right-click lane = event, plus • marker {time:F3}s selected when a marker is selected.

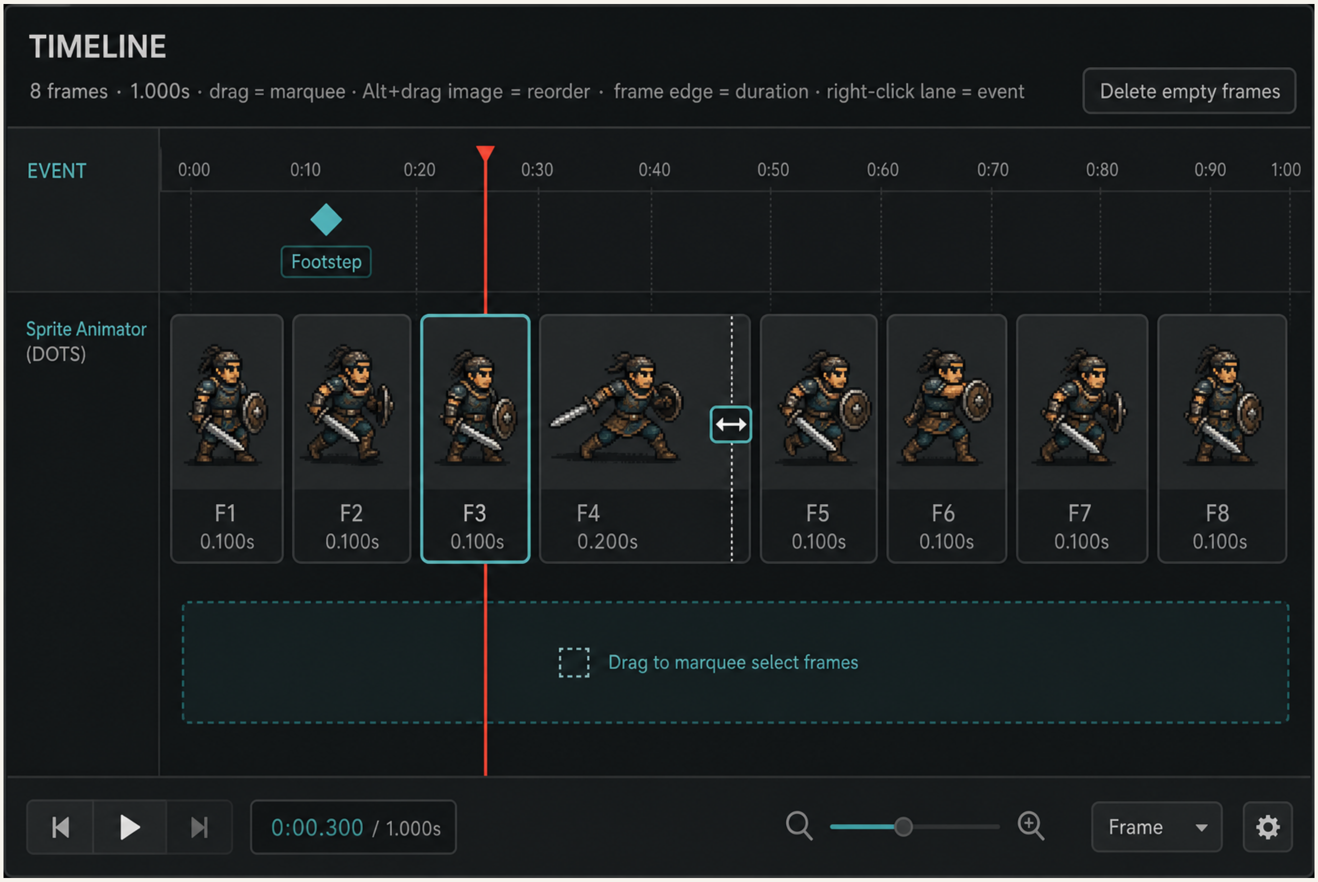


Figure 4.

TIMELINE pane of v0.7.1: ruler, EVENT lane with diamond markers, frame cards with sheet-cell thumbnails, red playhead, right-edge duration handle, and Delete empty frames. Left-drag marquee selects cards; Alt+drag the thumbnail image reorders the play list. Labeled illustration of the v0.7.1 editor — not a live Unity screenshot.

Control	Type / default	What it does
Playhead	Red 2 px needle + downward triangle	Continuous function of preview time across Loop / Once / Ping Pong / Reverse Loop (SpriteAnimPlayback.PlayheadX). Drag the needle, the ruler (y ≤ 27), or the EVENT lane to scrub. Scrub pauses playback. During Scrub / Reorder / Resize / Event drags the needle follows the pointer.
Left-drag marquee	TimelineDragMode.Marquee · card lane y > 54	Left-drag in the card lane, including starting on a card, begins a marquee. Shift / Ctrl / Cmd makes it additive. Cards whose rect overlaps join the selection. Starting on a right-edge handle and moving mostly vertically converts resize into marquee. Escape restores the baseline selection.
Alt+drag reorder	Alt+left-drag the thumbnail image (not the whole card)	Cursor MoveArrow. Hit shape is Circle by default, or the authored Polygon (TIMELINE INPUT). Ghost card: “Move frame {n}” / “release to place”. Commits clip.MoveFrame, which moves all per-frame metadata including socket FrameIndex. Moves under 3 px are ignored.

Control	Type / default	What it does
Right edge duration	2 px handle · cursor ResizeHorizontal	Drag horizontally: FrameDurationScales[i] = duration × FrameRate. Minimum while dragging: 0.02 s (inspector field minimum 0.001 s). Status: Frame {n} hold: {duration:F3}s. Undo: Change Frame Duration (discrete, committed on mouse-up).
Delete / Backspace	Global shortcut, timeline as last priority	With no collider, socket, or event selected, removes the selected frame(s). A clip must keep at least one frame (“A clip must keep at least one frame”).
Delete empty frames	Mini-button, top-right of the pane	Disabled when the clip has ≤ 1 frame or zero empty cells. Empty = sheet cell alpha ≤ 8. Tooltip when none: “No sheet cells in this clip are empty of opaque pixels.” Always keeps at least one frame.
EVENT lane	Label EVENT · y 27–54	Right-click (x ≥ 48) opens Add Event Marker/{name}, Add Event Marker/New Event..., and Clear Event Marker. Left-drag a diamond moves it in time (clamped to the clip). One Event ID per frame; 0 = none. Time is normalized 0...1 inside the frame (0 = frame start).
Middle-mouse / leftover drag	Horizontal pan	Pans the timeline. Scroll wheel also pans (delta.y × 32). Click on empty ruler/lane with no drag places the playhead.
Escape	Cancels in-progress timeline drag	Resize restores original duration; marquee restores previous frame selection.

Help box in TIMELINE INPUT (exact): “Circle is the default thumbnail hit target. Drag a thumbnail to reorder; drag empty track space to pan; drag the ruler or red playhead to scrub.” Onion skin is not drawn on the timeline — it is a preview overlay. Default profile events if the list is empty: Id 1 Name Footstep Color (0.35, 0.85, 1, 1); Id 2 Name Attack Color (1, 0.35, 0.3).

8 · INSPECTOR · PLAYBACK

The inspector is a scroll view (minimum content height about 1640 px + 26 px per collider row) under the header INSPECTOR. Groups are always-visible SectionLabel headers — no foldouts. PrepareInspectorUndo records “Edit Sprite Animator” on mouse-down / key-down / drag-perform. Figure 5 is the labeled inspector illustration for this and the next several chapters.

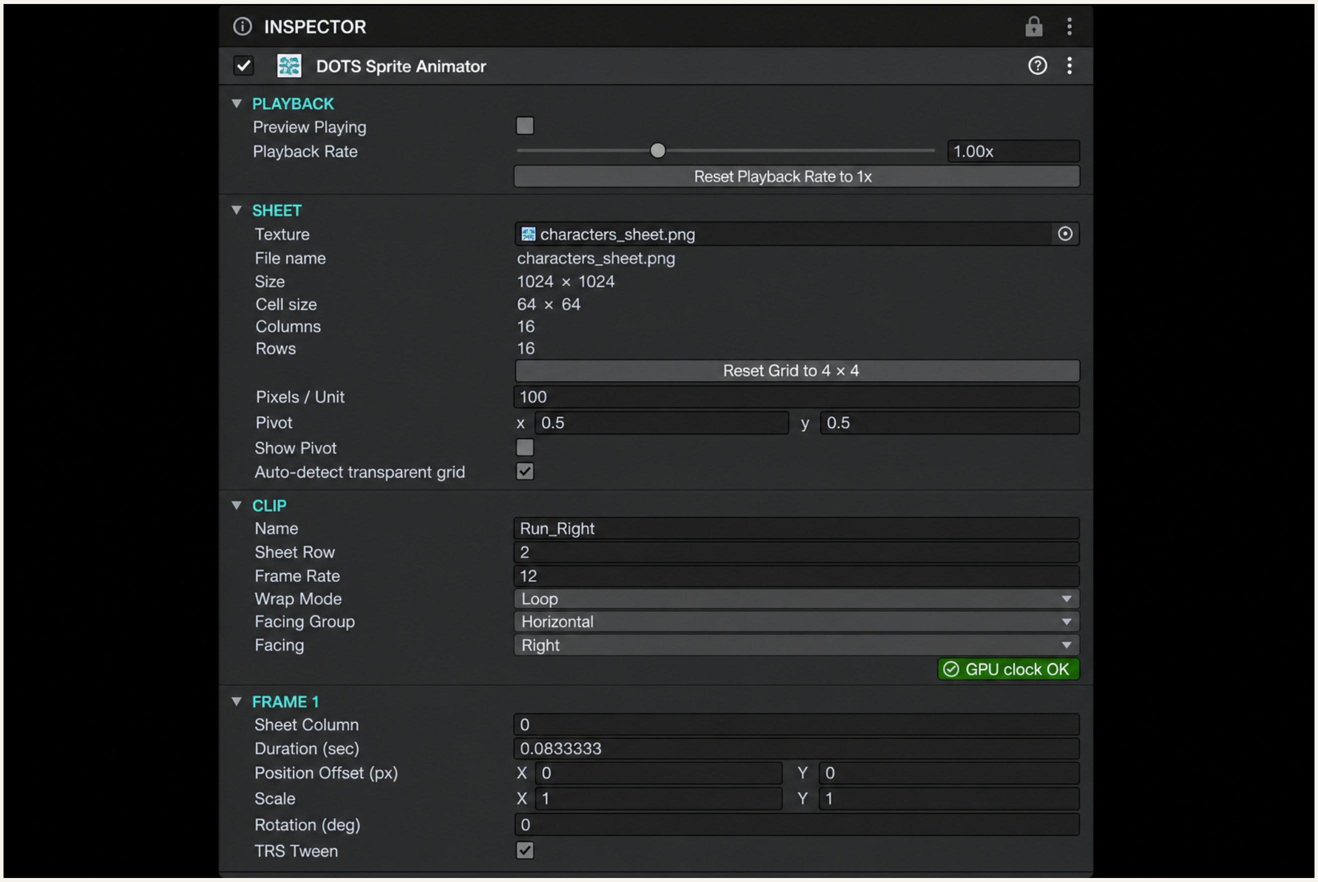


Figure 5.

INSPECTOR pane of v0.7.1 with always-visible SectionLabel groups: PLAYBACK, SHEET, TIMELINE INPUT, CLIP (including the GPU clock OK / CPU only badge), FRAME, then EVENT MARKER, SOCKETS, ONION SKIN, COLLIDER CREATION, and COLLIDERS further down the scroll. Labeled illustration of the v0.7.1 editor — not a live Unity screenshot.

PLAYBACK fields

Control	Type / default	What it does
Preview Playing	Toggle · default false	Same flag as toolbar Play/Pause. Preview is paused on window open.
Playback Rate	Slider · default 1 · range 0.1–3	Same value as toolbar Speed. Preview time advances by delta × max(0.05, speed).
Reset Playback Rate to 1x	Button · always visible	Sets speed to 1. Unlike toolbar 1x, this button does not disable when already 1.

9 · INSPECTOR · SHEET

There is no texture Filter Mode, importer PPU, wrap, mipmap, or compression field in this section. Pixels / Unit is the profile’s own scale used when baking offsets and sockets to world units.

Control	Type / default	What it does
Texture	ObjectField Texture2D · default null	Spritesheet. Assigning a texture auto-loads {SheetName}_profile.asset beside it if present (TryLoadExistingAsset). Save Profile stays disabled until this is set.
File name	LabelField · read-only · “—” without a texture	Texture asset filename, else texture name.
Size	LabelField · read-only · “—” without a texture	{width} × {height} pixels.
Cell size	LabelField · read-only · hidden until a texture	{width/Columns} × {height/Rows} integer division.
Columns	IntField · default 4 · min 1	Grid columns, left-to-right. DefaultColumns = 4.
Rows	IntField · default 4 · min 1	Grid rows, top-to-bottom. DefaultRows = 4.
Reset Grid to 4 × 4	Button · disabled at 4×4	Tooltip: “Reset the sheet grid to 4 columns by 4 rows.” Status: Reset sheet grid to 4 x 4.
Pixels / Unit	FloatField · default 100 · min 0.01	Converts authored pixel offsets and sockets to world units at bake (offset / PPU).
Reset (beside PPU)	Button 48 px · disabled at 100	Tooltip: “Reset Pixels Per Unit to 100.”
Pivot	Vector2Field · default (0.5, 0.5)	Normalized cell pivot. The green preview disc clamps 0–1 while dragging; the field itself is free.
Reset (beside Pivot)	Button 48 px · disabled at center	Tooltip: “Reset the pivot to centered (0.5, 0.5).”
Show Pivot	Toggle · width 92 · default true	Tooltip: “Draw the sheet pivot as a green dot in the preview. Drag it to move.”
Auto-detect transparent grid	Button · needs a texture	Counts opaque bands (alpha > 8) to set Columns × Rows. Status: Detected {c} × {r} grid, or No transparent gaps detected; set grid manually.

10 · INSPECTOR · TIMELINE INPUT

These fields author the hit-test shape used when Alt-dragging a timeline thumbnail to reorder. They do not change how frames look — only where the pointer has to land. Exact enum popup label: Thumbnail Hit Shape (Circle / Polygon).

Control	Type / default	What it does
Thumbnail Hit Shape	EnumPopup SpriteTimelineHitShape · default Circle (0)	Circle or Polygon. Circle is the default thumbnail hit target.
Reset (beside the enum)	Button 48 px · disabled at Circle + 8-point default polygon	Tooltip: “Restore circular thumbnail hit-testing and the default 8-point polygon.”
Polygon Vertices	IntSlider · default 8 · range 3–16	Visible only when shape is Polygon. Rebuilds a regular polygon.
Point {i}	Vector2Field × N · regular octagon around 0.5	Each axis clamped 0…1. Normalized points inside the thumbnail.
Reset Regular Polygon	Button	Tooltip: “Replace edited polygon points with an evenly spaced regular polygon.”

> **Help box (exact)**
> Circle is the default thumbnail hit target. Drag a thumbnail to reorder; drag empty track space to pan; drag the ruler or red playhead to scrub.

11 · INSPECTOR · CLIP

Requires a selected clip. There is no Tint or Flip on this inspector — Tint lives on SpriteAnimSetAuthoring.Tint (default white); Flip is a baked runtime component, not authored here. The GPU / CPU badge is a colored bar, not a field.

Control	Type / default	What it does
Name	DelayedTextField · + Clip uses Clip {n}; data default Idle	Clip identity. Runtime SpriteAnims.Play(..., name) matches this via FNV1a64. Hitboxes retarget on change.
Sheet Row	IntField · + Clip uses count % Rows · data default 0 · range 0 … Rows−1	Which sheet row (0 = top). Combined with each frame’s Sheet Column to pick a cell.
Frame Rate	FloatField · default 8 · min 0.1	Frames per second. Actual frame seconds = FrameDurationScales[i] / Frame Rate.
Reset (fps)	Button · disabled at 8	Tooltip: “Reset the clip frame rate to 8 fps.”
Wrap Mode	Popup · default Loop (byte 0)	Loop, Once, Ping Pong, Reverse Loop. Ping Pong / Reverse Loop force CPU only.
Reset (wrap)	Button · disabled at Loop	Tooltip: “Reset playback wrapping to Loop.”
Facing Group	DelayedTextField · default empty string	Tooltip: “Optional logical group name (e.g. Walk, Idle).” Used by SpriteAnims.PlayFacing.
Facing	EnumPopup SpriteFacingDirection · default None (255)	None / Right / UpRight / Up / UpLeft / Left / DownLeft / Down / DownRight. Tooltip: “Direction variant inside the facing group.”
GPU clock OK	Green badge + help “GPU clock OK.”	Uniform sequential Loop/Once clip: no offsets, custom holds, events, reorder, non-1 scale, rotation, non-Linear tween, or sockets.

Control	Type / default	What it does
CPU only	Red badge + first failing reason	See chapter 22 for the exact reason strings. SpriteGpuAnimSwitch.ToGpu refuses non-eligible clips.

12 · INSPECTOR · FRAME

Header is FRAME {_selectedFrame+1}, or FRAME {n} • {k} selected when multi-selected. Edits apply to the primary selected frame index. Duration (sec) = scale / fps (stored as FrameDurationScales[i] = duration × FrameRate; default scale 1).

Control	Type / default	What it does
Sheet Column	IntField · sequential 0…Columns−1 · range 0 … Columns−1	Column of this play-order slot. Reordering frames changes play order, not this column, unless you edit it. Non-sequential columns mark the clip CPU-only.
Duration (sec)	FloatField · default 1 / FrameRate seconds (scale 1) · min 0.001	Hold time. Disabled while a timeline resize drag is live. Custom holds imply CPU only. Undo from the timeline edge: Change Frame Duration.
Reset (duration)	Button · disabled at scale 1	Tooltip: “Reset this frame to one normal frame interval.”
Position Offset (px)	Vector2Field · default (0, 0)	Tooltip: “Per-frame position offset in source pixels (baked to runtime).” Data field name OnionOffsets (compat). Non-zero implies CPU only. Same data as onion Playback Offset (px).
Scale	Vector2Field · default (1, 1)	Tooltip: “Per-frame local scale multiplier.” Non-1 implies CPU only.
Rotation (deg)	FloatField · default 0	Tooltip: “Per-frame local z rotation in degrees.” Non-zero implies CPU only.
TRS Tween	EnumPopup SpriteEaseMode · default Linear	Linear / SmoothStep / EaseIn / EaseOut / Step. Tooltip: “Easing from this frame's TRS key to the next frame.” Non-Linear implies CPU only. Evaluate: Linear = t; SmoothStep = $t^2(3-2t)$; EaseIn = t^2; EaseOut = $1-(1-t)^2$; Step = 0 until t==1.
+ Frame After	Button	Inserts a frame after the primary selection. New column = min(Columns−1, currentColumn+1). Copies duration; offset 0, scale 1, rotation 0, tween Linear, no event. Shifts later sockets/hitboxes.
Remove Frame / Remove Frames	Button · disabled if only 1 frame	Label is Remove Frames when more than one frame is selected. Always keeps at least 1 frame.

13 · INSPECTOR · EVENT MARKER

Frame events are byte IDs with an exact in-frame timestamp. Existing profiles migrate to frame-start timing (0.0). One id per frame; 0 = none.

Control	Type / default	What it does
Frame	LabelField · “{n} of {count}”	Frame being inspected: selected marker frame, else the selected timeline frame.
Event ID	IntField · default 0 (none) · range 0–255	0 = no marker. Non-zero writes EventIds[frame] and creates a SpriteEventDef if missing (Event {id}).
(help when id = 0)	HelpBox	Exact: “No marker on this frame. Enter an Event ID, or right-click the timeline event lane at an exact time.”
Time (sec)	FloatField · default frame start (normalized 0)	Exact timestamp, clamped to that frame’s authored span. Stored as normalized 0…1 inside the frame.
Event Name	DelayedTextField · Footstep / Attack / Event {id}	Human-readable label for the byte id. Shared across the profile.

Control	Type / default	What it does
Event Color	ColorField	Diamond color on the EVENT lane. Footstep cyan, Attack red, HSV for newly created ids.
Selected: {name}	Muted label	Current definition name.
Delete Marker	Button · width 104	Clears EventIds[frame] to 0.

14 · INSPECTOR · SOCKETS

Section header: SOCKETS — FRAME {n}. Count line: {n} socket(s) • {k} selected. The same Name is one identity across frames; Offset X/Y and Angle are per-frame keys. Missing keys fall back to the last known pose. Sockets stay CPU-only and do not tween at runtime.

Control	Type / default	What it does
Add Socket / Click Preview to Place…	Button · cyan when armed · default disarmed	Arms click-to-place. Second click on the button cancels (Socket placement cancelled). Armed help: “Click on the frame to place a socket. Escape or right-click cancels.” Empty help: “Add Socket, then click the preview to place a named attach point.”
{i}:{name}	Mini-button row + color chip	Select that identity. Shift add, Ctrl/Cmd toggle. Off-frame rows show “ (other frame)” and “No key on this frame yet. Drag or edit to add one.”
Name	DelayedTextField · Socket / Socket {n}	Renames the identity on every frame. Empty becomes Socket.
Offset X (px)	FloatField · default 0	FrameSocketDef.LocalPosition.x (+x right, source pixels).
Offset Y (px)	FloatField · default 0	LocalPosition.y (+y up).
Angle (deg)	FloatField · default 0	LocalAngle. No socket tween at runtime.
Clear Frame Offset	Button	Tooltip: “Reset this frame's socket position and angle to 0. Adds a key if this frame did not have one.”
Delete	Button	Tooltip: “Delete this socket identity from every frame.”

15 · INSPECTOR · ONION SKIN

Onion is a preview overlay (past ghosts cooler/left, future warmer/right). It is not drawn on the timeline. Click a badge to select a layer; drag or arrow-nudge to author that frame’s Position Offset (px). Arrow keys on a selected ghost preempt timeline frame-step.

Control	Type / default	What it does
Enabled	Toggle · default false (OnionSkinEnabled)	Draws ghost frames in the preview. Disabling clears onion selection.
Past Frames (Left)	IntField · default 3 · range 0 … Frames−1	How many earlier play-order frames to ghost.
Future Frames (Right)	IntField · default 3 · range 0 … Frames−1	How many later frames to ghost.
All Past	Button	Sets past count to Frames−1.
All Future	Button	Sets future count to Frames−1.
Show Layer Numbers	Toggle · default true	Draws −{n} / +{n} badges.
Playback Preview	EnumPopup PreviewOffsetMode · default Authored	Authored / Centered. Same as preview View: Offsets / View: Centered.
Playback Offset (px)	Vector2Field · that frame’s OnionOffsets	Shown when a visible ghost is selected (label: Selected ghost {±d} • frame {n}). Same data as FRAME Position Offset (px).
Recenter Selected	Button · disabled at (0, 0)	Tooltip: “Reset this onion ghost offset to (0, 0).” Right-click the selected ghost does the same.

Control	Type / default	What it does
Reset Onion Defaults	Button · disabled at 3 / 3 + numbers on	Tooltip: “Restore 3 past frames, 3 future frames, and visible layer numbers.”

Unselected help (exact): “Click a ghost to select its frame. Drag or use arrow keys to align it; the source-pixel offset is baked into runtime playback.” Shift+arrow nudges 5 px; arrow alone nudges 1 px.

16 · INSPECTOR · COLLIDER CREATION

Muted instruction: Select a shape before placing it. Creation tools own the preview canvas and suppress transform handles until cancelled.

Control	Type / default	What it does
Show Colliders	Toggle · default true	Master preview visibility. Off cancels creation, clears selection, and stops drafts. Hidden colliders still save and bake.
Square / Circle / Polygon	Segmented toggles · none armed by default	SpriteColliderShape. Idle tooltip: “Arm the {Shape} collider creation tool.” Active: “{Shape} creation is armed. Click again to cancel.” Square/Circle: click for a default-size collider (about 12% of the shorter cell side) or drag from center to size. Polygon: click vertices (3–16); click first point, double-click, or Enter closes; Right-click/Backspace removes last; Escape cancels.
Continuous Placement	Toggle · default false	Keep the shape tool armed after each place.
New Collider ID	IntField · default 1 · range 1–255	Gameplay id written to FrameBoxDef.Id (byte).
Cancel Creation Tool	Button · disabled when no tool is armed	Status: Collider creation cancelled. Right-click on the canvas also cancels Square/Circle.

Help boxes (exact, by mode)

None: Choose a shape to create, or click existing colliders to select them. Drag empty preview space for marquee selection.
Polygon: Click to draw polygon vertices. Click the first point, double-click, or press Enter to close. Right-click/Backspace removes the last point; Escape cancels.
Square / Circle: Click for a default-size collider or drag from its center to size it. Right-click cancels the active creation tool.

17 · INSPECTOR · COLLIDERS

Section header: COLLIDERS — FRAME {n}. Count line: {n} collider(s) • {k} selected. Empty help: This frame has no colliders. Arm a shape above, then click the preview. Preview handles (chapter 6) edit the same records. Hidden colliders still bake.

Control	Type / default	What it does
Show / Hide (eye)	Per-row mini-button · Unity visibility icons, fallback text Hide / Show	Toggles box.Hidden. Tooltips: “Hide this collider in the preview” / “Show this collider in the preview.” Undo: Hide/Show Sprite Collider. Hidden colliders are not drawn unless selected (reduced alpha); they still bake.
{i}. {Shape} • ID {id}	Mini-button · appends “ (hidden)” when hidden	Select this collider. Tooltip: “Select this collider. Shift adds; Ctrl/Cmd toggles.”
×	Mini-button · tooltip “Delete this collider.”	Deletes that collider.
Angle (deg)	FloatField · default 0 · shown when a primary selection exists	Tooltip: “Rotation around the collider center. Saved on the profile.” Mirrors the rotate handle.
Reset (angle)	Button 48 px · disabled at 0	Tooltip: “Reset this collider's rotation to 0°.”
Select All	Button	Selects every collider on this frame.
Delete Selected	Button · disabled with empty selection	Deletes the current collider selection.

Control	Type / default	What it does
Copy to Next Frame	Button · disabled on the last frame	Tooltip: “Clone current-frame colliders onto the next frame.”
Copy to All Frames	Button	Tooltip: “Clone current-frame colliders onto every other frame in this clip.”
Delete All on This Frame	Button	Tooltip: “Delete every collider on the current frame. This action supports Undo.”

Preview context menu on a collider: Delete {Shape} Collider (or Delete Selected Colliders ({n})), Select All Colliders on Frame, Delete All Colliders on Frame. RectUV is top-left origin in cell UV; Angle is degrees around the rect center (authoring y-down). Hidden is editor-only.

18 · History List panel

List is the 0.7.1 addition. It sits beside Redo in the toolbar. The label flips to Hide while the panel is open. This overlay does not replace Unity’s own Undo window.

Control	Type / default	What it does
List / Hide	Toolbar button · tooltip “Show the undo/redo history panel.”	Toggles the history overlay. New in 0.7.1.
Undo / Redo (window title)	Floating GUI.Window id 99221	Default rect (40, 56, 280, 340); repositions to window.width − 300, 52 if too small.
Title-bar drag	GUI.DragWindow on the top 20 px	Drag the title bar to move the overlay around the studio.
Undo / Redo	Buttons inside the panel	Undo.PerformUndo / PerformRedo — same as the toolbar.
Done	Bold header	Scroll list of named undo operations, newest first. Current head marked with a pointer. Empty: “No actions yet.”
Redo (section)	Bold header · only if the redo stack is non-empty	Lists redo names under Done.

Names are pushed only for discrete operations (PushUndoName). The catch-all inspector group “Edit Sprite Animator” is not listed. Named examples: Add Sprite Animation Clip, Rename Sprite Animation Clip, Change Frame Duration, Move Sprite Pivot, Add Sprite Animation Event, Move Sprite Socket, Rotate Sprite Collider, Nudge Onion Skin Offset.

19 · Keyboard shortcuts

Handled in HandleGlobalShortcuts (plus a few local handlers). Shortcuts are ignored while a named string field is focused (clip rename or delayed text fields), except as noted. Do not document clip-list reorder keys, collider arrow-nudge, Ctrl+D, Ctrl+S, or frame-copy shortcuts — they are not implemented.

Control	Type / default	What it does
Space	Play / pause	Toggles preview playback. Ignored while editing a string field. Releases keyboard focus first. Debounced 80 ms because KeyCode.Space and character == space can arrive as a pair. No-op with no clip. Status: Playback started / Playback paused. Consumed only while this window has focus.
F2	Rename selected clip	The only clip-list rename binding. No double-click, no Enter-to-rename.
Delete / Backspace	Priority delete	(1) Backspace during polygon draft removes last vertex; (2) delete selected colliders and sockets; (3) delete selected event marker; (4) remove selected timeline frames. Cannot delete the last remaining frame.
Ctrl+Z / Cmd+Z	Undo	Unity Undo. Not locally bound; toolbar tooltip documents it. Button calls Undo.PerformUndo.
Ctrl+Y or Ctrl/Cmd+Shift+Z	Redo	Unity Redo. Documented on the Redo button.
Left / Right arrows	Step one frame (pauses)	Only if no onion ghost is selected. When a ghost is selected, arrows nudge that ghost 1 px (Shift = 5 px) instead.
Ctrl+A / Cmd+A	Select all preview objects	All colliders (if Show Colliders) and all sockets on the current frame. Does not select timeline frames.
Alt+drag thumbnail	Reorder frames	Timeline only. Alt+left-drag on the preview pans instead — same modifier, different pane.

Control	Type / default	What it does
Alt+left-drag preview / MMB	Pan preview	See chapter 6.
Shift+click / Ctrl+click	Additive / toggle selection	Frames, colliders, and sockets (timeline, preview, inspector lists).
Escape	Cancel / clear	Timeline drag; collider transform restore; then clears collider/socket/event/onion selection, creation tool, socket tool, pivot drag, marquee. Polygon draft: cancels the whole tool. Status when clearing: Selection and active tools cleared.
Enter / keypad Enter	Commit or close	Commits clip rename, or closes a polygon draft with at least 3 vertices.
Mouse wheel	Zoom or pan	Preview zoom / timeline horizontal pan, depending on the pane under the cursor.

20 · Save and load profiles

A profile is a ScriptableSpriteSheetProfile (ScriptableObject wrapper around SpriteSheetProfile Data) plus a pretty JSON twin for diffing. Inspector edits call SetDirty but do not write JSON until Save Profile.

Control	Type / default	What it does
Save Profile	Toolbar button · disabled without a Texture	Writes, in the texture's folder: <SheetName>_profile.asset and <SheetName>_profile.json. If no asset is bound yet, reuses a sibling with that name or CreateAsset. Notification: Profile saved. Without a sheet: Assign a sprite sheet before saving.
Load Profile…	Toolbar button · native file picker	Project-relative .asset that is a ScriptableSpriteSheetProfile. Failures: Selected profile must be inside this Unity project / Selected file is not a ScriptableSpriteSheetProfile.
Assign Texture	Auto-load	If {name}_profile.asset already sits beside the texture, it loads automatically.
Select asset + open window	OnEnable / LoadAsset	Select the profile in the Project window, then open or focus the studio.
New Profile	Discards the asset binding	Blank in-memory SpriteSheetProfile: 4×4, PPU 100, pivot center, onion off, two default event defs, no clips, no hitboxes.

File names (exact)
Beside the spritesheet: <SheetName>_profile.asset (ScriptableSpriteSheetProfile) and <SheetName>_profile.json (pretty JsonUtility of SpriteSheetProfile). Example: hero_profile.asset + hero_profile.json.

21 · Runtime playback and events

Add SpriteAnimSetAuthoring (InvertLab.Sprites.DOTS, DisallowMultipleComponent) to a GameObject, assign the profile, and bake (SubScene or conversion). The baker prefers Profile when set (it overrides Sheet, grid, and Clips). DependsOn(profile) and DependsOn(sheet).

Authoring component (Unity inspector, not the studio)

Control	Type / default	What it does
Profile	ScriptableSpriteSheetProfile · optional	Tooltip: "Optional profile authored in Window > DOTS Sprite Animator. When set, it overrides Sheet, grid, and Clips below."
Sheet	Texture2D	Used only if Profile is unset. Tooltip: "Spritesheet: grid of frames, left-to-right then top-to-bottom".
Columns / Rows	int · default 4 / 4	Used only if Profile is unset.
Clips[]	ClipAuthoring array	Fallback clip list. Sample defaults: Idle 8 fps loop, Run 10 fps loop, Attack 12 fps once, Block 8 fps loop.
InitialClipIndex	int · default 0	First clip to play; baked into SpriteAnimPlayer.ClipIndex.
SizeUnits	float · default 1 · min 0.01	World size. Not shown in the studio.
Tint	Color · default white	Baked as SpriteTint. Not a studio clip field.

Bake produces SpriteAnimSetRef (blob) + SpriteAnimPlayer + SpriteAnimFrame + SpriteTint + SpriteAnimEnabled + SpriteFlip + SpriteAnimEventBuffer + SpriteAnimEventsPending + SpriteSocketBuffer. Rendering is wired by SpriteAnimRenderInitSystem; ticking by SpriteAnimPlayerSystem. Offsets and sockets from a profile are divided by Pixels / Unit.

Gameplay API

From QuickStart.md. Play matches clip Name via FNV1a64 and restarts at t = 0 (Playing = 1). ReverseLoop starts on the last frame. PlayFacing matches Facing Group + Facing and falls back to the first clip in that group.

```
SpriteAnims.Play(entityManager, entity, "Run");
SpriteAnims.PlayFacing(entityManager, entity, "Walk", SpriteFacingDirection.Down);
```

README also documents the eight-way form:

```
SpriteAnims.PlayFacing(entityManager, entity, "Walk", SpriteFacingDirection.DownLeft);
```

Events after SpriteAnimPlayerSystem

From Documentation~/AnimationEvents.md. The player emits an event when playback crosses a marker, including every marker crossed during a long update. Receivers must run after SpriteAnimPlayerSystem. Only entities with events are matched because SpriteAnimEventsPending is enableable. Events remain readable through the rest of the simulation tick; the package clears them immediately before animation playback on the next tick.

```
[UpdateInGroup(typeof(SimulationSystemGroup))]
[UpdateAfter(typeof(SpriteAnimPlayerSystem))]
public partial struct FootstepSystem : ISystem
{
    public void OnUpdate(ref SystemState state)
    {
        foreach (var (events, entity) in
                 SystemAPI.Query<DynamicBuffer<SpriteAnimEventBuffer>>()
                          .WithAll<SpriteAnimEventsPending>()
                          .WithEntityAccess())
        {
            for (int i = 0; i < events.Length; i++)
                Handle(entity, events[i].Id, events[i].ClipIndex, events[i].FrameIndex);
        }
    }
}
```

Managed bridge (MonoBehaviours and other managed systems):

```
void OnEnable()  => SpriteAnimEvents.Raised += OnAnimationEvent;
void OnDisable() => SpriteAnimEvents.Raised -= OnAnimationEvent;

void OnAnimationEvent(Entity entity, SpriteAnimEventBuffer evt)
{
    if (evt.Id == 1)
        PlayFootstep(entity);
}
```

Programmatically-created animator entities can call SpriteAnimEvents.Ensure(entityManager, entity). Authoring and factory paths install the required components automatically.

22 · GPU clock vs CPU-only eligibility

Each clip shows a colored badge in INSPECTOR · CLIP. GPU clock OK (green) means the compact GPU clock can drive a uniform sequential Loop/Once clip. CPU only (red) means the CPU player must run. SpriteGpuAnimSwitch.ToGpu(...) refuses non-eligible clips. Sockets never GPU-animate.

Control	Type / default	What it does
GPU clock OK	Green bar + help “GPU clock OK.”	Wrap Mode is Loop or Once; frames are sequential sheet columns starting at Frames[0]; every duration scale is 1; no events; offsets (0,0); scale (1,1); rotation 0; TRS Tween Linear; no sockets.
CPU only	Red bar + first failing reason	The badge prints the first reason from SpriteGpuEligibility.IsGpuEligible (see table below).

Reason strings (exact)

Reason (help box)	Triggered by
No clip loaded.	Clip is null or has no frames.
Ping-pong and reverse playback require CPU timing.	Wrap Mode is Ping Pong or Reverse Loop.
Sockets require CPU playback.	clip.Sockets.Count > 0. Sockets have no GPU tween.
Frame reorder requires CPU playback.	A play-order column is not Frames[0] + index.
Per-frame offsets require CPU playback.	Any Position Offset (px) / OnionOffsets is not (0, 0).
Custom frame holds require CPU playback.	Any FrameDurationScales[i] is not 1.
Animation events require CPU playback.	Any Event ID is not 0.
Per-frame scale requires CPU playback.	Any Scale is not (1, 1).
Per-frame rotation requires CPU playback.	Any Rotation (deg) is not 0.
TRS easing requires CPU playback.	Any TRS Tween is not Linear.

Authoring implication
Keep idle and run loops GPU-eligible when you can (Loop or Once, sequential columns, uniform holds, no events/sockets/TRS). Put footsteps, smear offsets, facing-socket weapons, and ping-pong recoveries on CPU-only clips — the badge will say so immediately.

Invert Lab · DOTS Sprite Animator · User Guide v0.7.1 · 27 August 2026 · Unity 6000.5. Editor header PackageVersion = 0.7.1; package.json version = 0.7.0. Window > DOTS Sprite Animator · com.invertlab.spriteanimator · InvertLab.Sprites.DOTS.